Colin Michael Vargas

M.S. CANDIDATE

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Education

Arizona State University Tempe, AZ
M.S. Aerospace Engineering (Propulsion/Combustion) Expected May 2027

Arizona State University Tempe, AZ
B.S.E. Aerospace Engineering (Aeronautics), Minor in French May 2025

Technical Experience

Arizona State University Tempe, AZ
Grader / Instructional Assistant, Advanced Math Methods for Engineers Jan 2025 - Present

- Guided students in interpolation, numerical integration, approximation, and linear algebra methods through one-on-one technical support and structured solution development.
- Evaluated submitted work for mathematical correctness and numerical reasoning, translating method selection and solution logic into concise written and in-person feedback.

Honeywell Aerospace Phoenix, AZ
In-Service Engineering Intern May 2024 - Aug 2024

- Supported avionics and propulsion-related engineering investigations by organizing case data, emulator outputs, and technical documentation to strengthen follow-up analysis and troubleshooting continuity.
- Consolidated artifacts across open customer cases to improve traceability, accelerate engineering handoffs, and preserve investigation context across multi-step technical workflows.

Selected Projects

M.S. Thesis Research
Compressible CFD, AMR, & Parallel Scientific Computing 2025 - Present

- Developing C++ solver infrastructure for compressible CFD with long-term application to multiphase aerospace flow and thermo-fluid simulation.
- Studying adaptive mesh refinement workflows in AMReX to improve fidelity-to-cost tradeoffs in high-resolution flow modeling.
- Implementing finite-volume methods in parallel using AMReX to support scalable development of compressible thermo-fluid simulations.

Parametric Ramjet / Scramjet Analysis Tool
MATLAB, Thermodynamic Modeling, & Technical Report Fall 2024

- Built an analytical propulsion model in MATLAB to estimate thrust, overall efficiency, thrust-specific fuel consumption, and specific impulse across multiple operating conditions.
- Applied thermodynamic and combustion relationships to compare ramjet and scramjet design conditions and document propulsion performance tradeoffs in a technical report.
- Performed parametric studies over flight Mach number, altitude, and inlet/nozzle conditions to identify operating regimes that maximize thrust, efficiency, and fuel economy.

2D Navier–Stokes Mixing Chamber Simulation
MATLAB, ParaView, & Technical Report Spring 2025

- Modeled the conservative 2D Navier–Stokes equations for a mixing chamber, separating convective and diffusive terms to advance the solution numerically in time.
- Generated velocity and temperature field visualizations to compare Reynolds- and Prandtl-number effects on mixing behavior and transport-driven flow development.
- Used staggered-mesh finite-difference methods, pressure projection, and grid-convergence analysis to evaluate how Reynolds and Prandtl number variations altered transient mixing and heat-transfer behavior.

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programming C++, Python, MATLAB • **simulation** AMReX, ParaView, XFOIL
development Git, CMake, Linux, Command Line, LaTeX • **languages** English, French